

BRAINBOTS · PROJECT GUIDE

Smart Attendance System

How the whole project works, explained in plain language for anyone — no technical background needed.

Card reader · Cloud messaging · Server · Database · Web dashboard
Prepared September 2026

CONTENTS

What's inside

1	The project in one page	What it is and why it exists
2	The big picture	The six parts and how they connect
3	Each part, explained simply	Reader, cloud, server, database, dashboard
4	What happens when a card is tapped	The full journey of one tap
5	In time and out time rules	How the system decides in, out or ignore
6	When the internet goes down	Offline working and catching up
7	Adding a new student or staff member	Card enrolment step by step
8	ID numbers	How student and staff IDs are made
9	Who can sign in	The three kinds of admin accounts
10	A tour of the dashboard	Every screen and what it shows
11	The staff side	Departments, staff and their attendance
12	Reports (PDF)	Printing attendance for a student, batch or everyone
13	What information is kept	The records, in plain words
14	Safety and security	Passwords, privacy and good practice
15	Setting it up and running it	For the person who looks after it
16	Limits and future ideas	What it does not do yet
17	Common questions	Quick answers
18	Glossary	Technical words made simple

PART 1

The project in one page

A card-tap attendance system for the BrainBots centre. Students and staff tap an ID card at the door, and their arrival and leaving times are recorded automatically and shown live on a website.

The problem it solves

Taking attendance by hand — a paper register or calling out names — is slow, easy to get wrong, and hard to search later. It also cannot tell you when someone arrived or left, or how long they stayed.

What this system does instead

- Every student and staff member gets a small **ID card** (like a bus pass or office access card).
- At the door there is a small **card reader**. A person simply holds their card near it.
- The reader instantly shows a **green light and one beep** (accepted) or a **red light and three beeps** (not registered).
- The first tap is the **in time**. A later tap is the **out time**. The time spent is worked out automatically.
- The admin sees everything on a **web dashboard** — on a phone, tablet or computer — updated the moment a card is tapped.
- Attendance can be downloaded as a neat **PDF report** for one student, a whole batch, or everyone.
- It keeps working even when the **internet is down**, and catches up automatically when it comes back.

In one sentence: Tap a card at the door → the system notes who came, when they came and when they left → the admin can see it live and print reports whenever needed.

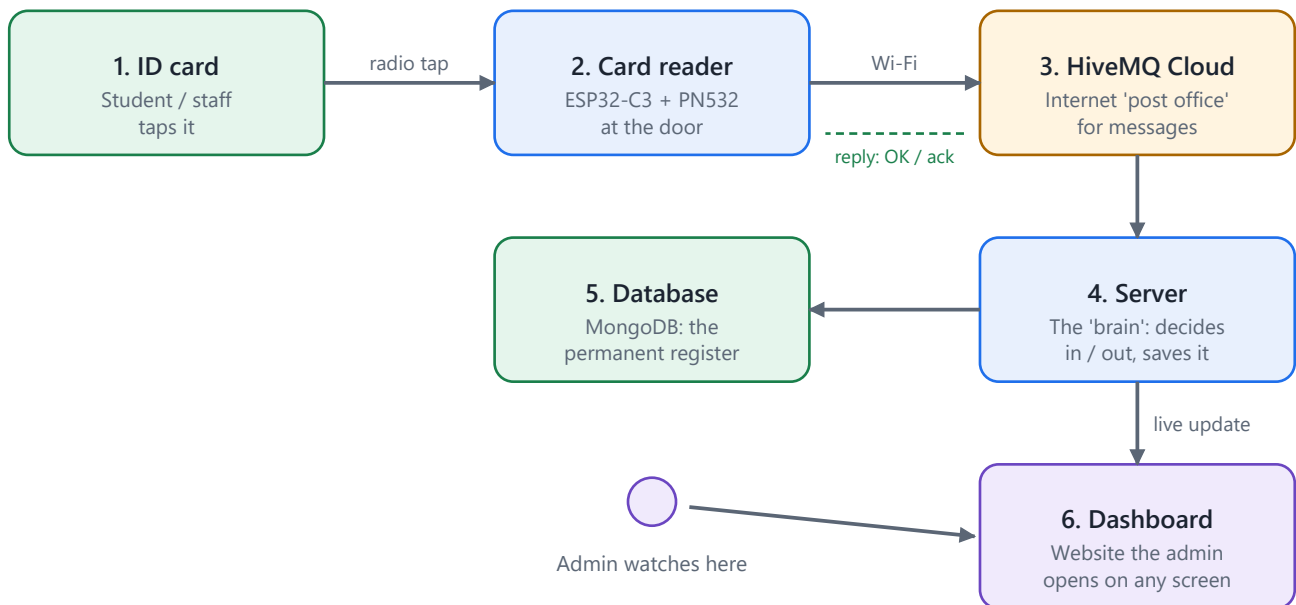
Who it is for

Person	What they do with it
Student	Taps their card when arriving at and leaving each class. Nothing else to do.
Staff member	Taps their card when arriving and leaving, the same way.
Student admin	Watches student attendance, adds and edits students and batches, prints reports.
Staff admin	Does the same for staff members and departments.
Super admin	Can see and manage both the student side and the staff side.

PART 2

The big picture

The system has six parts. Think of it like a letter travelling from the door of the centre to the office.



#	Part	Everyday comparison	Its job
1	ID card	A key	Carries a unique number. Every card's number is different.
2	Card reader	The security guard at the door	Reads the card, decides on the spot, lights up and beeps, and passes the news on.
3	HiveMQ Cloud	A post office on the internet	Carries short messages between the reader and the server, in both directions.
4	Server	The office clerk	Works out in time and out time, applies the rules, and tells the dashboard.
5	Database	The register book / filing cabinet	Stores every student, staff member, batch and attendance record permanently.
6	Dashboard	The notice board	The website the admin opens to see attendance and manage everything.

The reader never talks to the website directly, and the website never talks to the reader directly. Everything passes through the server, just as all letters go through the office.

PART 3

Each part, explained simply

1 · The ID card

Each card has a tiny chip with a serial number burned in at the factory. The system only uses this number — nothing else is written on the card. A card on its own means nothing; it only becomes "Priya's card" when the admin links it to Priya in the system.

2 · The card reader (the box at the door)

The reader is a small home-made device built from three main pieces:

- **ESP32-C3 SuperMini** — a tiny computer, about the size of a thumb, with Wi-Fi built in. It is the reader's brain.
- **PN532** — the card-reading board. It picks up the card's number when the card is held near it.
- **A green light, a red light and a buzzer** — to give the person instant feedback.

The important part: the reader keeps its **own copy of the list of registered cards** in its memory. So when a card is tapped, it does not need to ask anyone — it decides immediately (in well under a second). It also keeps a notebook of taps that it has not yet been able to send, so nothing is lost if the internet stops.

3 · HiveMQ Cloud (the messenger)

This is an online service that passes small messages back and forth, a bit like a WhatsApp group between the reader and the server. The reader posts "card A42B9F01 was tapped at 9:00" and the server replies "got it". The server also posts the up-to-date list of registered cards so the reader always has a fresh copy. The connection is encrypted so outsiders cannot read it.

4 · The server (the brain)

A program running on a computer at the centre (or online). It receives each tap, looks up whose card it is, applies the in/out rules (Part 5), saves the result in the database and instantly pushes the news to anyone who has the dashboard open. It also makes the PDF reports and checks passwords when someone signs in.

5 · The database (the register)

Where everything is stored permanently: the batches, the students, the staff, the departments and every single attendance record. It uses a program called **MongoDB**. Even if the server is switched off, the data stays safe here.

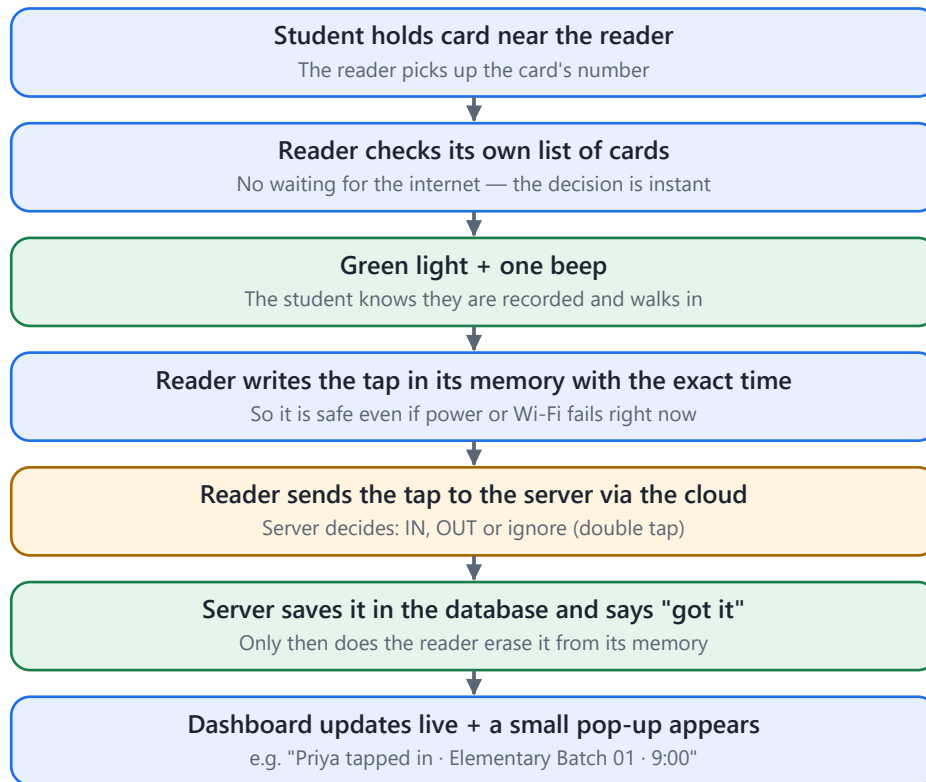
6 · The dashboard (the website)

A modern website that opens in any browser. The admin signs in with a username and password and sees live attendance, who is in, who has not come, and can add students, edit batches and print reports. It works on phones and computers and switches to a dark theme automatically if the device uses one.

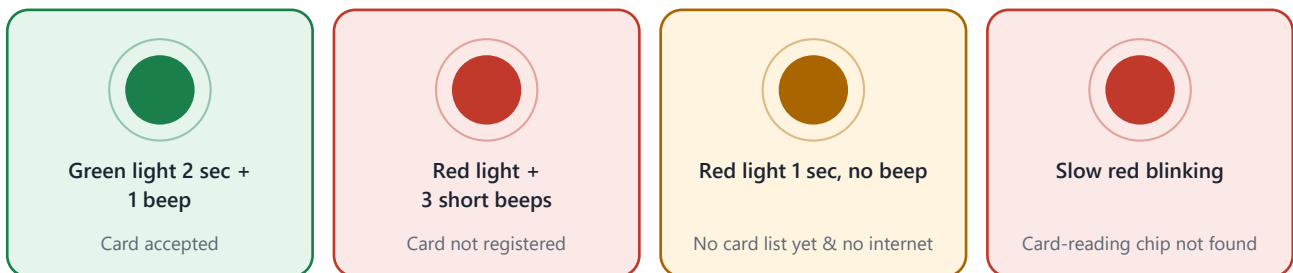
PART 4

What happens when a card is tapped

Here is the full journey of one tap, from the student's hand to the admin's screen. It usually takes about a second.



What the lights and beeps mean



Situation	What the reader does	Is it recorded?
Card is registered	Green light 2 seconds, one short beep	Yes — as in, out, or ignored double tap
Card is not on the reader's list, internet working	Asks the server. If the server also doesn't know it: red light, three beeps	No (the admin sees an "unknown card" pop-up)
Card not on the list, no internet	Red light, three beeps	Saved and checked later, in case the list was out of date

Situation	What the reader does	Is it recorded?
Reader has never received a card list and there is no internet	Red light 1 second, no beeps	Saved and checked later

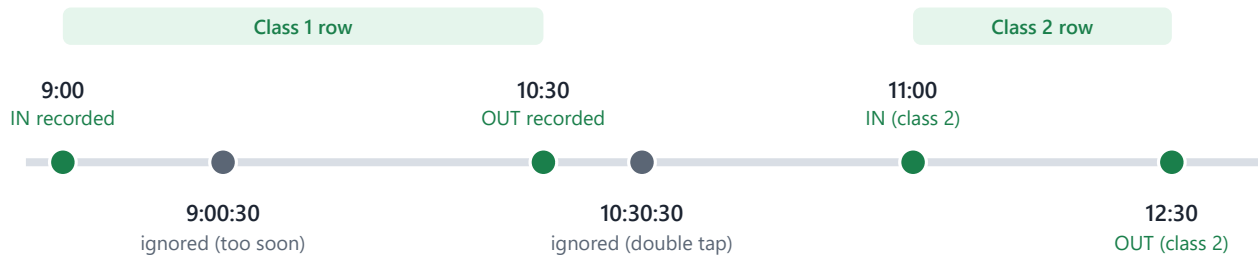
The special "red, no beep" signal exists so that a network problem is never mistaken for a rejected card.

PART 5

In time and out time rules

A batch may have several classes in one day. Each class becomes one row in the register with an in time and an out time. The server uses a few simple rules to decide what each tap means.

Example: one student's day



The rules

When the tap happens...	What the system does
No class is currently open for this student	Starts a new row — this tap is the in time
Too soon after the in time (under the minimum gap)	Ignored as an accidental double tap
After the minimum gap, and within 12 hours of the in time	Sets the out time — the class is closed
Less than 60 seconds after an out time	Ignored as a double tap
60 seconds or more after an out time	Starts a new row — the in time of the next class
More than 12 hours after an in time with no out	Starts a new row ; the old one shows a dash (–) for out

The **"minimum gap"** is a setting. The standard value is 10 minutes, but on this installation it has been set to **1 minute** so students can tap in and out quickly. The trade-off: a card brushed twice more than a minute apart will count as an out tap.

Good to know

- Every valid card **always gets the green light**, even if that tap is ignored as a double tap. The card is valid; only the record is skipped.
- When a tap is ignored, the dashboard shows a small pop-up ("tapped again too soon — ignored"), so the admin knows why nothing changed.
- If a student forgets to tap out of one class, their next tap will be taken as that class's out time.

- "Time on site" (duration) is simply out time minus in time, calculated automatically.
- All times are kept in world time (UTC) but shown and grouped by the centre's local time zone (Asia/Colombo), so a late-evening tap is filed under the correct date.
- Staff taps follow exactly the same rules.

PART 6

When the internet goes down

One of the strongest features: the reader does not stop working when Wi-Fi or the internet fails.

How it keeps working

1. The reader already has its own copy of every registered card number, so it can still say green or red instantly.
2. Each tap is written into the reader's memory with the **real time** it happened. It can hold up to **400 taps**.
3. When the internet comes back, the reader sends the stored taps to the server, **oldest first**.
4. The server files each tap **as if it had just happened at its original time**, so the in and out times are exactly right.
5. After the server confirms each tap, the reader deletes it from its memory.

Safety checks built in

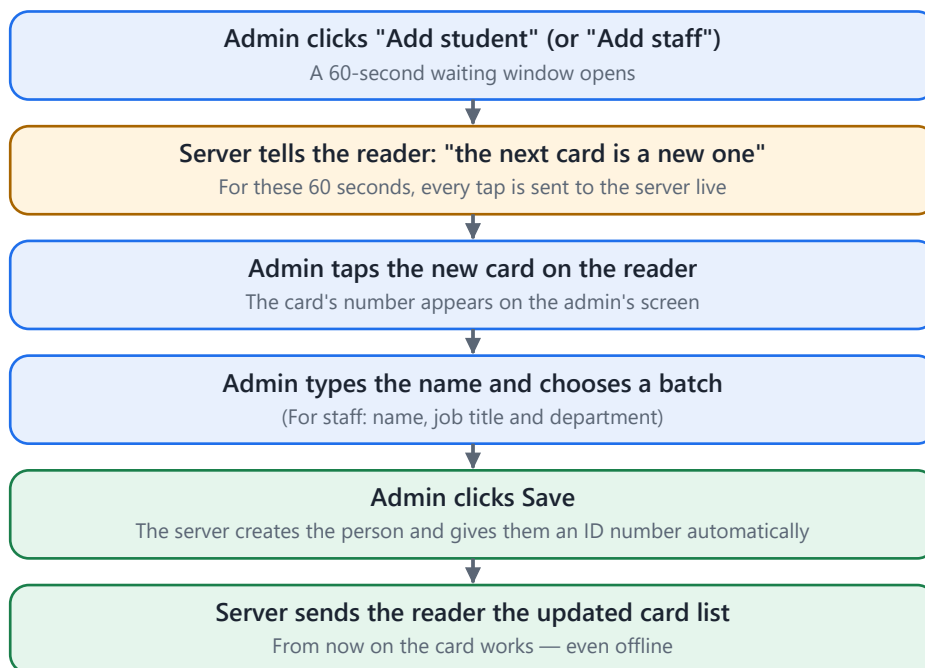
- **No double counting:** every stored tap has a unique tag. If the same tap is accidentally sent twice, it is only recorded once.
- **Power cuts:** the card list and stored taps are kept in memory that survives the reader being switched off.
- **Keeping the clock right:** the reader gets the correct time from the internet (or from the server). It also saves the time every 10 minutes. If it restarts without internet, it estimates the time and marks those taps as approximate, then corrects them as soon as the real time is known.
- **Card list updates:** whenever a student or staff member is added or removed, the server sends the reader a fresh list (card numbers only — never names).

What does NOT work offline: adding a new card (enrolment) needs the reader to be online, and the dashboard only shows offline taps once the reader has reconnected and sent them.

PART 7

Adding a new student or staff member

Linking a new card to a person is called enrolment. It takes under a minute and is done from the dashboard.



What if something goes wrong?

Situation	What happens
The card already belongs to someone	The reader shows red, and the dashboard says who owns that card
No card is tapped within 60 seconds	The window closes and the dashboard says it timed out; just try again
The reader is offline	Enrolment cannot work — the dashboard shows the reader's connection status

PART 8

ID numbers

Every student and staff member is given a permanent ID number automatically. Nobody has to invent or type one.

Student IDs

Bb / EL / 26 / 0101

Part	Meaning	Example
Bb	Centre code (BrainBots)	Bb
EL / IL / AL	Level: Elementary, Intermediate or Advanced	EL
26	Intake year (2026)	26
0101	Running number for that level, starting at 0101	0101, 0102, ... 0199, 0200 ...

- All batches of the same level share one number sequence. So Elementary Batch 02's first student might be 0117 if Batch 01 already used 0101–0116.
- IDs **never change** — not even if a student moves to another batch or level.
- Two admins adding students at the same moment will never get the same number.

Staff IDs

Staff get a similar ID in the form **Bb/ST/0101**, with one running sequence for all staff.

PART 9

Who can sign in

Only signed-in admins can see or change anything. Students and staff never need to sign in — they only tap cards.

Account type	Can see and manage	Lands on
Admin (student side)	Batches, students, student attendance, student reports	Batches page
Staff admin	Departments, staff members, staff attendance, staff reports	Departments page
Super admin	Everything on both sides	Batches page (with Staff tabs too)

- Accounts are created once by a setup script. There is no "sign up" button — nobody from outside can create an account.
- Passwords are never stored as plain text; they are scrambled (hashed) so even someone who sees the database cannot read them.
- A sign-in lasts 12 hours, after which the admin signs in again.
- The admin's username and password can be changed at any time with one command (see Part 15).

PART 10

A tour of the dashboard

What each screen shows and what you can do on it.

Screen	What you see	What you can do
Front page	A public introduction page explaining the system, with a sample picture of the dashboard	Click Sign in
Sign in	Username and password boxes	Sign in to the dashboard
Batches (home)	Three summary tiles — In today , Non-attendees , Students — and a card for every batch, grouped by level, with a live count of who is in	Click a tile to see the actual list of names · Add, rename or delete a batch · Make a Full report
Batch page	One batch's attendance: every student's in time, out time and time on site	Switch between Total students , Not came and In class · Pick Today / Yesterday / Last 7 days or any dates · Search a name · Make a Batch report
Students	The full list of students with their ID, name, batch and card	Search and filter · Add student (tap card) · Edit name or batch · Delete · Make a student Report
Student history	One student's full record, grouped day by day	Choose a period · Make a PDF report
Departments	Staff version of the Batches page: In today, Non-attendees, Staff tiles and a card per department	Add, rename or delete departments · Staff report
Department page	One department's staff attendance	Same filters as the batch page
Staff	The list of staff members	Add staff (tap card), edit, delete, reports
Staff history	One staff member's full record	Choose a period · PDF report

Live pop-ups

While the dashboard is open, small pop-up messages appear on **any page** the moment something happens:

- Someone tapped **in** or **out** (with name, batch and time).
- Someone tapped **again too soon** and it was ignored.
- An **unknown card** was tapped.
- The **reader went offline or came back online**.

Nothing needs refreshing. Counts, lists and tiles update by themselves as cards are tapped.

Batch rules worth knowing

- A batch that still has students in it **cannot be deleted**. Move or remove its students first (from the Students page).
- Renaming a batch or changing its level does not change any student's ID.
- Old attendance records always remember the batch the student was in at that time, even if they move later.

PART 11

The staff side

Staff use the same card reader at the same door. The system automatically knows whether a card belongs to a student or a staff member.

Student side	Staff side (the equivalent)
Batch (e.g. Elementary Batch 01)	Department (e.g. Teachers, Office)
Student	Staff member (with a job title, e.g. "Robotics teacher")
Student ID Bb/EL/26/0101	Staff ID Bb/ST/0101
Admin account	Staff admin account
Batches, Students pages	Departments, Staff pages

- Tap rules are identical: first tap signs in, a later tap signs out, and a tap after that signs in again.
- The student admin never sees staff information and the staff admin never sees student information. Only the super admin sees both.
- Two departments (Teachers and Office) are created at setup; more can be added from the dashboard.

PART 12

Reports (PDF)

Attendance can be turned into a clean, printable PDF document in two clicks.

Kinds of report

Report	Where to find it	What it contains
One student	Students page → Report, or the student's history page	Name, ID, batch, period, and a table: Date · In · Out · Duration, plus total visits
One batch	Batch page → Batch report	Every student in the batch: Date · Student ID · Name · In · Out · Duration
All batches	Batches page → Full report	Everyone, with an extra Batch column (printed sideways/landscape)
Staff	Staff side pages	The same, for staff and departments

How to make one

1. Click the report button.
2. Pick a period: **Today**, **Last week**, **Last 30 days**, **All time**, or choose your own dates. In the Full report you can also pick a single batch from a drop-down.
3. Click **Generate PDF**. The PDF opens in a new browser tab, where it can be saved or printed.
4. If your browser blocks the new tab, click **Download PDF** instead.

Tip: use "All time" when a student leaves or a batch finishes, to keep a complete record of their attendance.

PART 13

What information is kept

The system keeps only what it needs. Here is everything stored in the database, in plain words.

Record	What it holds
Admin accounts	Username, scrambled password, and account type (admin / staff admin / super admin)
Batches	Name, level (EL / IL / AL) and display order
Students	Student ID, name, card number, and which batch they are in
Student attendance	Which student, which batch at the time, the date, in time and out time
Departments	Name and display order
Staff	Staff ID, name, job title, card number and department
Staff attendance	Which staff member, department, date, in time and out time
ID counters	The last number used for each level, so the next ID is always unique
Processed taps	A short-term memory of taps already recorded (kept 30 days), to stop duplicates

Important: deleting a student also deletes all their attendance records. This cannot be undone — download an "All time" report first if the history might be needed.

PART 14

Safety and security

- **Sign-in required:** all attendance and personal information is behind a username and password.
- **Separate sides:** the student admin and staff admin each see only their own side.
- **Scrambled passwords:** passwords are stored hashed, never as readable text.
- **Encrypted messaging:** the reader and server talk over an encrypted (TLS) connection through HiveMQ Cloud.
- **Minimal data to the reader:** the reader only ever receives card numbers, never names.

Before going fully live — checklist

Item	Why
Change all default admin passwords	The setup uses simple starter passwords; anyone who knows them could sign in
Set a long random secret key (JWT_SECRET) in the settings file	It protects sign-in sessions from being forged
Turn on certificate checking in the reader (USE_TLS_VALIDATION = 1)	Currently off for easy setup; turning it on stops anyone pretending to be the cloud service
Keep the settings file (.env) and the reader config private	They contain the Wi-Fi and cloud passwords

PART 15

Setting it up and running it

This part is for the person who looks after the system. It is kept short — the README file in the project has full technical details.

What is needed

- A computer with **Node.js** (runs the server and website) and **MongoDB** (the database) installed.
- A free **HiveMQ Cloud** account (the messenger).
- The card reader, assembled and programmed using the **Arduino IDE**.
- Wi-Fi at the door where the reader sits.

Everyday commands (typed in the project folder)

Command	What it does
<code>npm run seed</code>	First time only: creates the starting batches, departments and admin accounts. Safe to run again.
<code>npm start</code>	Starts the server (the brain). Must be running for anything to be recorded.
<code>npm run client</code>	Starts the website. Then open <code>http://localhost:5173</code> in a browser.
<code>npm run set-admin -- NAME PASSWORD</code>	Changes an admin's username/password (at least 8 characters).
<code>npm test</code>	Runs automatic checks on the ID numbers and date/time-zone logic.

After changing any server file or setting, the server must be **stopped and started again** for the change to take effect.

Reader wiring (for reference)

Card-reading board / part	Connects to reader pin
VCC (power)	5V
GND (ground)	GND
SDA (data)	GPIO 4
SCL (clock)	GPIO 5
Green light (with 330 Ω resistor)	GPIO 6
Red light (with 330 Ω resistor)	GPIO 7

Card-reading board / part	Connects to reader pin
Buzzer	GPIO 10

The card-reading board's small switches must be set to **I2C mode**. Pins 2, 8 and 9 are deliberately left unused because they can stop the reader from starting. If the reader blinks red slowly after switching on, the card-reading board is not connected properly.

Settings you can change

Setting	Meaning	Current / default
OUT_MIN_MINUTES	Minimum minutes between in and out tap	1 here (standard 10)
REJOIN_MIN_SECONDS	Seconds after an out tap before a new in is allowed	60
VISIT_WINDOW_HOURS	After this long without an out tap, the row is closed	12
ENROLL_WINDOW_SECONDS	How long "Add student" waits for a card	60
TZ_NAME	Local time zone	Asia/Colombo
INTAKE_YEAR	Year used in new student IDs	26

PART 16

Limits and future ideas

Things the system does not do today. None of them are faults — they simply have not been built yet.

Not yet available	Current workaround / idea
Replacing a lost card for an existing student	Today: delete and re-add the student (this loses history and uses a new ID). A small change could allow swapping the card directly.
Export to Excel / CSV	Use the PDF reports for now.
Creating admin accounts from the dashboard	Accounts come from the setup script; passwords are changed with a command.
Different beep for in / out / ignored	All valid taps give the same green light and beep today.
Reordering batches by drag-and-drop	Batches are shown in the order they were created within each level.
Live refresh inside the name-list pop-up	Close and reopen the pop-up to see the latest list (the tiles behind it do update live).
Running several copies of the server at once	One server is enough for one centre.
Very large centres	The card list sent to the reader holds about 900 cards.

PART 17

Common questions

A student tapped but nothing changed on the dashboard. Why?

Most likely they tapped again too soon after their in tap, so it was ignored as a double tap. A pop-up saying "tapped again too soon" appears in this case. If the reader was offline, the tap will appear once it reconnects.

The reader shows red for a student who is registered.

The reader's card list may be out of date. When online, it will check with the server automatically. Make sure the reader has Wi-Fi.

The internet was down all morning. Is attendance lost?

No. The reader stored up to 400 taps with their real times and sends them when the internet returns. The times recorded are the real tap times.

Can a student tap for a friend?

The system records the card, not the person. Anyone holding a card can tap it — the same as with any card system.

Does moving a student to another batch change their ID or old records?

No. The ID stays the same, and old records keep showing the batch they were in at that time.

Why can't I delete a batch?

It still has students. Move them to another batch or delete them first.

A student forgot to tap out.

Their next tap (within 12 hours) becomes the out time. After 12 hours, the row is left with a dash for out time.

Can I use more than one reader (e.g. two doors)?

Yes. Each reader is given its own name in its settings file; everything else works the same.

Can I check attendance from my phone?

Yes. The dashboard works on phones, as long as the phone can reach the server's address.

PART 18

Glossary

Technical words used in this project, explained simply.

Word	Plain meaning
Card number (UID)	The unique serial number inside each ID card.
ESP32-C3	A tiny, cheap computer chip with Wi-Fi. The brain of the card reader.
PN532 / NFC / RFID	The technology that reads a card held close to the reader, without touching.
Firmware	The program that runs inside the card reader.
MQTT	A lightweight way of sending short messages between devices over the internet.
HiveMQ Cloud	The online service that delivers those messages (the 'post office').
Server / backend	The main program that receives taps, applies rules and saves data.
Node.js / Express	The tools the server is built with.
Database / MongoDB	Where all the information is permanently stored.
Dashboard / frontend	The website the admin uses. Built with a tool called React.
Live updates (Socket.IO)	The technology that pushes new taps to the open website instantly, without refreshing.
Enrolment	Linking a new card to a new student or staff member.
Roster / card list	The list of all registered card numbers kept inside the reader.
Offline queue	The reader's notebook of taps waiting to be sent when the internet returns.
Acknowledgement (ack)	The server's "got it" reply that lets the reader erase a stored tap.
NTP	An internet service that gives devices the exact current time.
UTC / time zone	UTC is world time; the system shows everything in the centre's local time.
TLS / encryption	Scrambling messages so outsiders cannot read them on the way.
Hashed password	A password turned into unreadable scrambled text before being stored.
Sign-in token (JWT)	A digital pass given after signing in, valid for 12 hours.
.env file	The private settings file holding passwords and options for the server.

Summary: Tap card → reader decides instantly (green / red) → message travels through the cloud → server applies in/out rules → saved in the database → admin sees it live and can print reports. Works offline, never double-counts, and keeps student and staff sides separate.